Project Idea – Automatic Railroad Crossing Gate

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This project involves creating an automated railroad crossing gate system that uses sensors and actuators to control gate operation for vehicle safety. Three pairs of IR transmitters and receivers, two servo motors, and one ultrasonic sensor are used. The first IR pair detects an approaching train and activates the servo motors to close the gates on both sides of the track, while the second IR pair detects when the train has passed to reopen them. The third IR pair works with the ultrasonic sensor to detect obstructions on the crossing; if the ultrasonic sensor senses an object within a minimum distance, it sends a warning to the cloud so a connected train can slow down or stop. When the third IR pair detects that a train is near, it disables the ultrasonic sensor to prevent false obstruction readings. There will also be flashing alternating red LEDs and a buzzer to alarm when the gates are being closed. The purpose of this system is to ensure safety for the cars crossing the intersection by warning them with lights and sounds and putting down the gates to block them from crossing.